

Mobile Application Systems and Design
Lab 5.

Q1. Dependencies used and why

1. sqflite: I used sqflite because it is the main Flutter plugin for working with SQLite databases. It allows the app to create tables and perform CRUD operations such as insert, query, update and delete.
2. Path: I used path to correctly build database file path. Since mobile devices may use different path formats, this package helps to safely combine the folder path and database file name.
3. Intl: I used intl to format dates and display time in a user-friendly way in the app.

* SQLite is necessary because the app manages posts without the internet so SQLite stores structured data directly on the device, so that the staff members can still create, read, edit and delete posts while offline. Flutter's official guidance presents SQLite as the local solution for structured data.

Q2. Handling database exceptions

a) Database not initialized.

I used a singleton DatabaseHelper class with a database getter. This checks whether the database already exists in the memory. If it does not, the app initializes it before running any query. This prevents null database errors.

b) Insert, update and delete errors

Each database method is placed inside a try-catch ~~error~~ ^{block}. If an operation fails, the app throws a meaningful exception such as "Insert failed" or "delete failed". The UI then displays the error using a snackbar or message on screen.

c, Invalid or corrupted data

When reading data from SQLite, I convert rows into a Post object using `Post.fromMap()`. In that method, I check whether required fields exists. If important fields are missing, I throw a `FormatException`. This helps detect corrupted or Invalid data.

Q3. SQLite in Flutter

a) Database vs Table.

A database is the whole container that stores data files. In this app, the database is `posts_manager.db`.

A table is a structure inside the database that stores records in rows and columns. In this app, the table is `posts` with columns:

- id
- title
- content
- author
- createdAt

So the database is the full storage system, while the table is one organized section inside it.

b) CRUD operations

CRUD means:

- Create: adding a new post using `insert`.
- Read: fetching all posts or one post using query
- Update: editing an existing post using `update`
- Delete: removing a post using `delete`

All these operations are supported directly ~~using~~ by SQLite.

c, How Flutter interacts with SQLite asynchronously

Flutter interacts with SQLite using `async` and `await`. Database operations can take time, so they run asynchronously instead of blocking the UI.

The `sqflite` package states that database operations run in a background thread on Android and iOS, which helps keep the interface smooth while data is being loaded / saved.

Example: `final posts = await DatabaseHelper.instance.getAllPosts();`

This means flutter waits for the database result without freezing the app screen.

Offline Posts Manager Testing and Screenshots

This document provides step-by-step test scenarios to demonstrate how the app works.

1. Test Scenario 1: Launching the App

Steps:

1. Run the Flutter app
2. Observe the home screen

Expected Result:

- App opens successfully
- AppBar displays "Offline Posts Manager."
- If no posts exist, an empty state message is shown

Offline Posts Manager



No posts yet.
Tap + to add your first post.

+ New Post



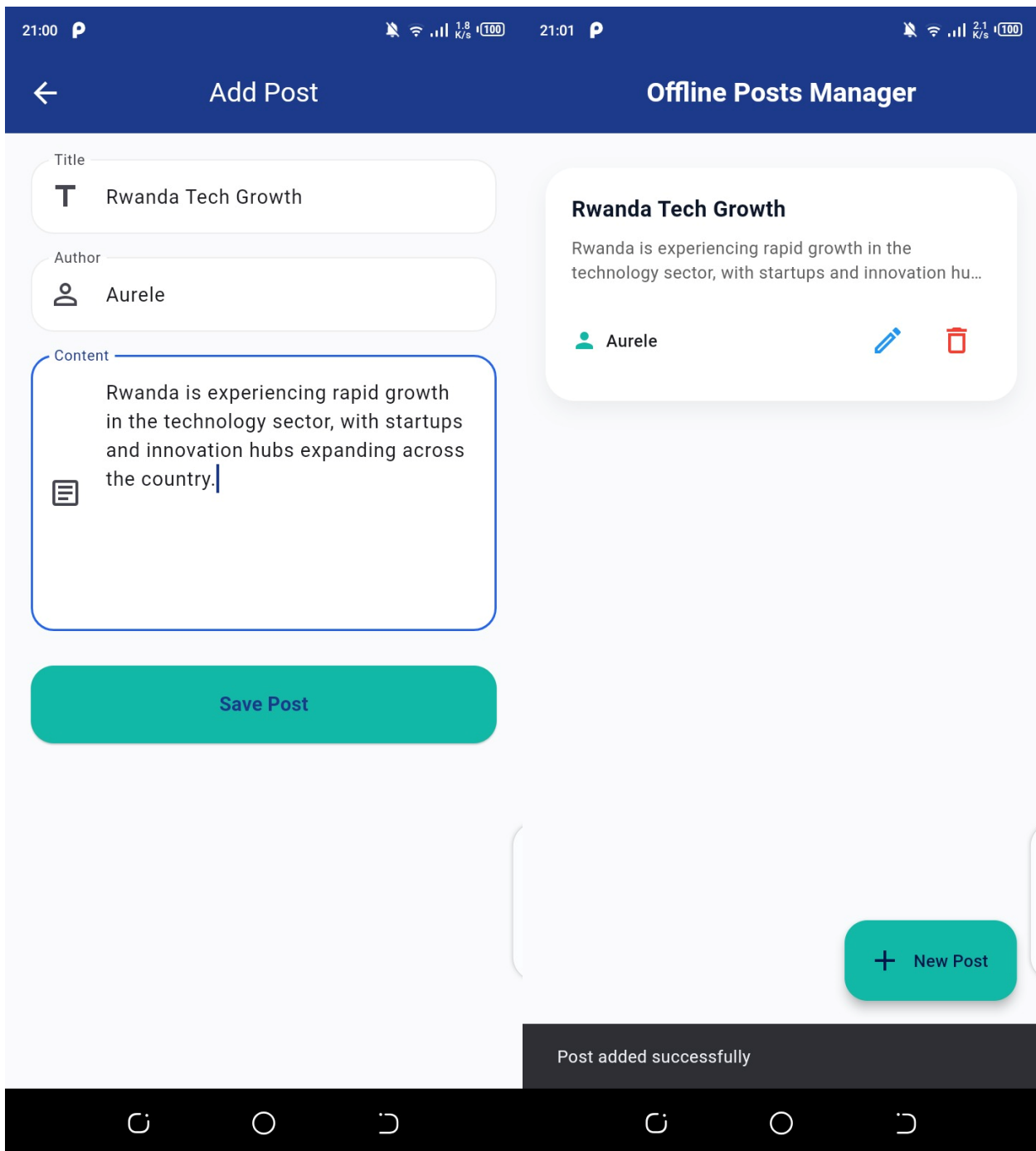
2. Test Scenario 2: Adding a New Post

Steps:

1. Click the "New Post" button
2. Enter the details
3. Click "Save Post"

Expected Result:

- Post is saved successfully
- Success message appears
- App returns to home screen
- New post appears in the list



4. Test Scenario 3: Viewing Post Details

Steps:

1. Tap on a post from the list

Expected Result:

- Details screen opens
- Displays full title, author, date, and content



Post Details

Rwanda Tech Growth



Aurele



2026-03-23 21:01

Content

Rwanda is experiencing rapid growth in the technology sector, with startups and innovation hubs expanding across the country.



5. Test Scenario 4: Adding Multiple Posts

Steps:

1. Add Post 2, Post 3, and Post 4

Expected Result:

- All posts appear in the list
- Posts are displayed in descending order (latest first)

Offline Posts Manager

Health System Improvements

The healthcare sector is integrating digital systems to improve patient management and service delivery.

 Diane



Education Digitalization

Schools are adopting digital tools and online learning platforms to enhance education accessibil...

 Eric Nshimiyimana



Agriculture Innovation

Smart farming techniques using IoT are improving crop yields and helping farmers monitor environme...

 Jean Bosco



Rwanda Tech Growth

Rwanda is experiencing rapid growth in its technology sector, with startups and digital innovation...

 New Post



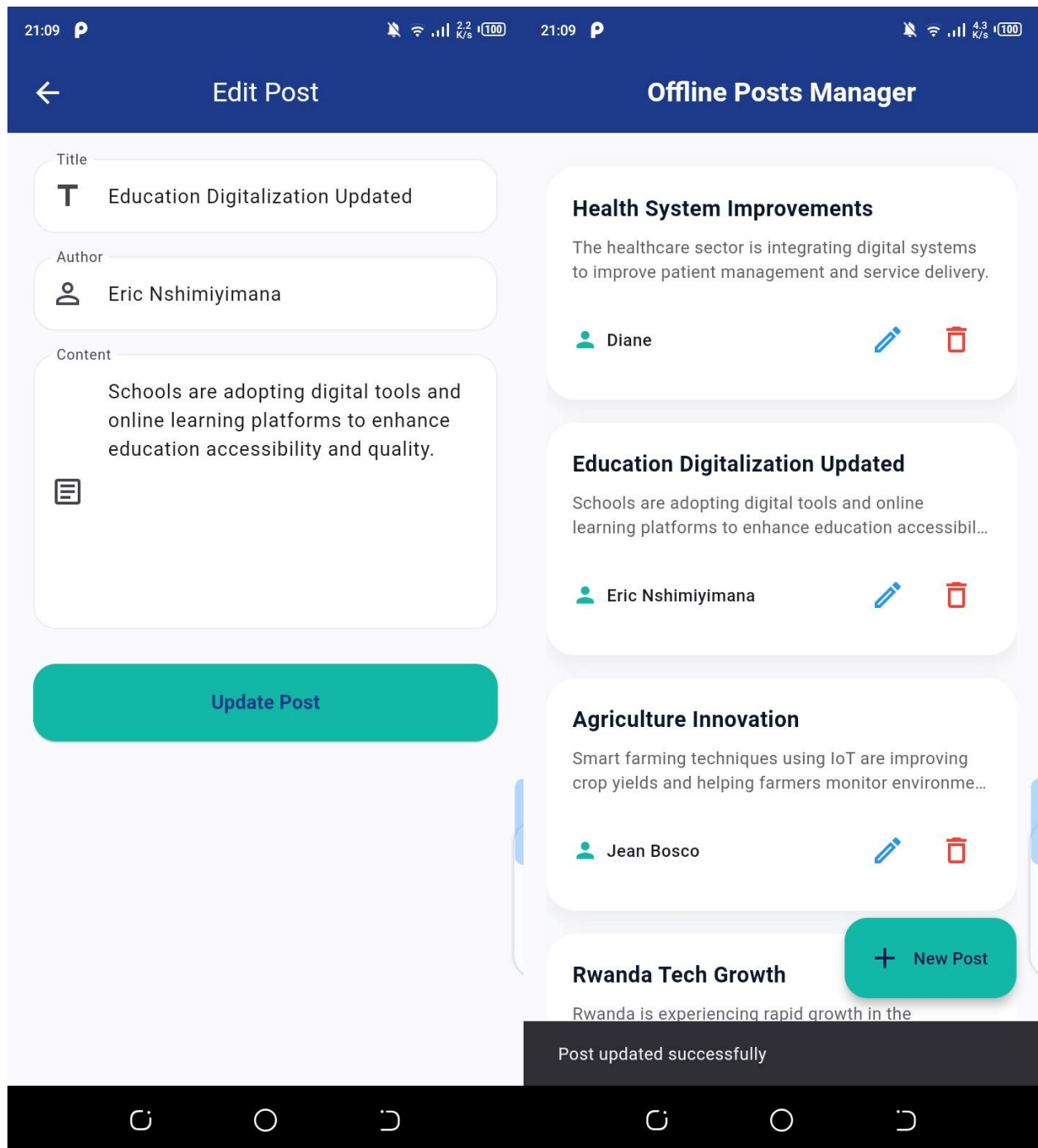
6. Test Scenario 5: Editing a Post

Steps:

1. Click the edit icon on a post
2. Modify the title to: "Education Digitalization Updated"
3. Click "Update Post"

Expected Result:

- Post is updated successfully
- Updated title appears in the list



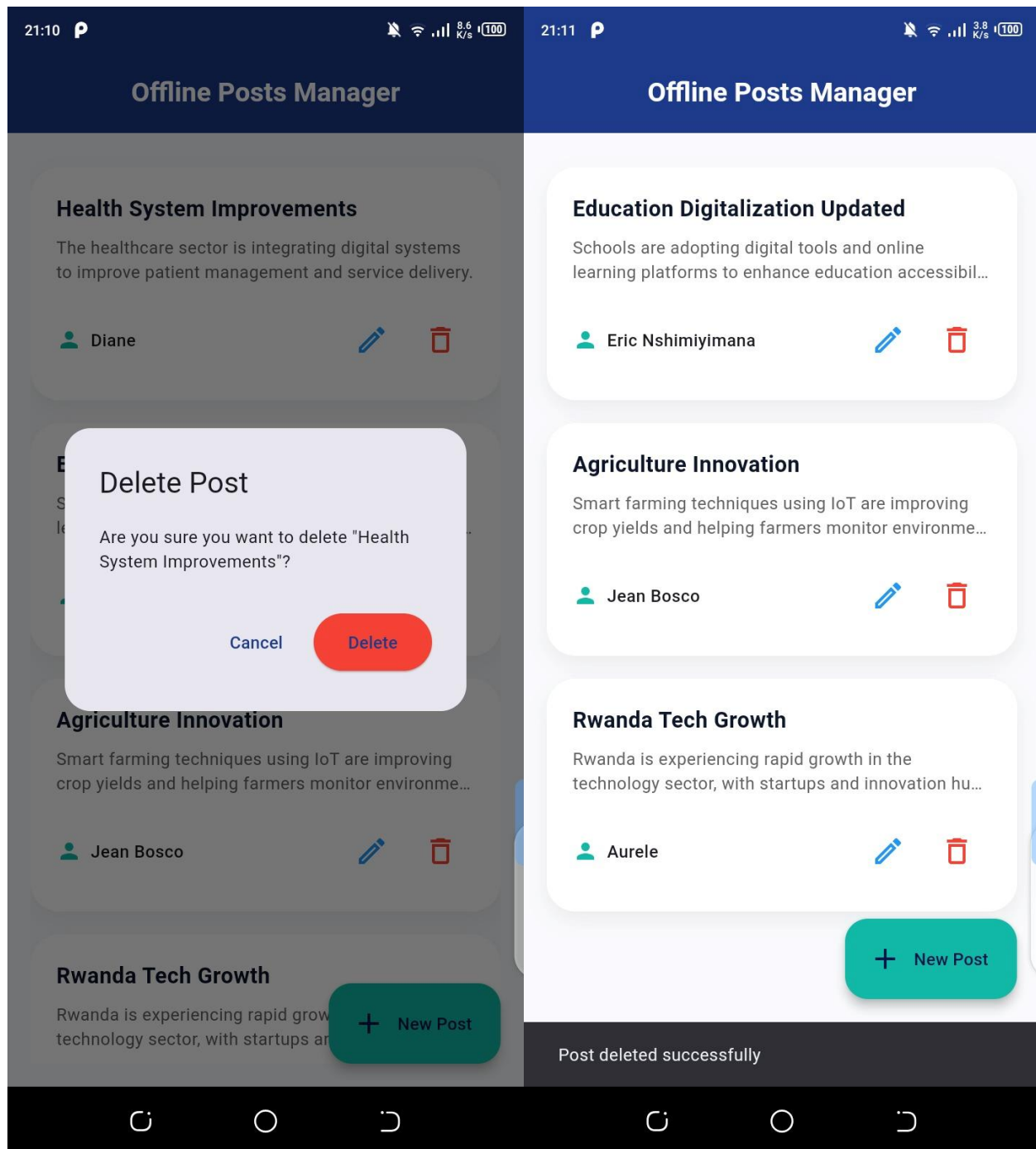
7. Test Scenario 6: Deleting a Post

Steps:

1. Click the delete icon on a post
2. Confirm deletion in the dialog

Expected Result:

- Post is removed from the list
- Confirmation message appears



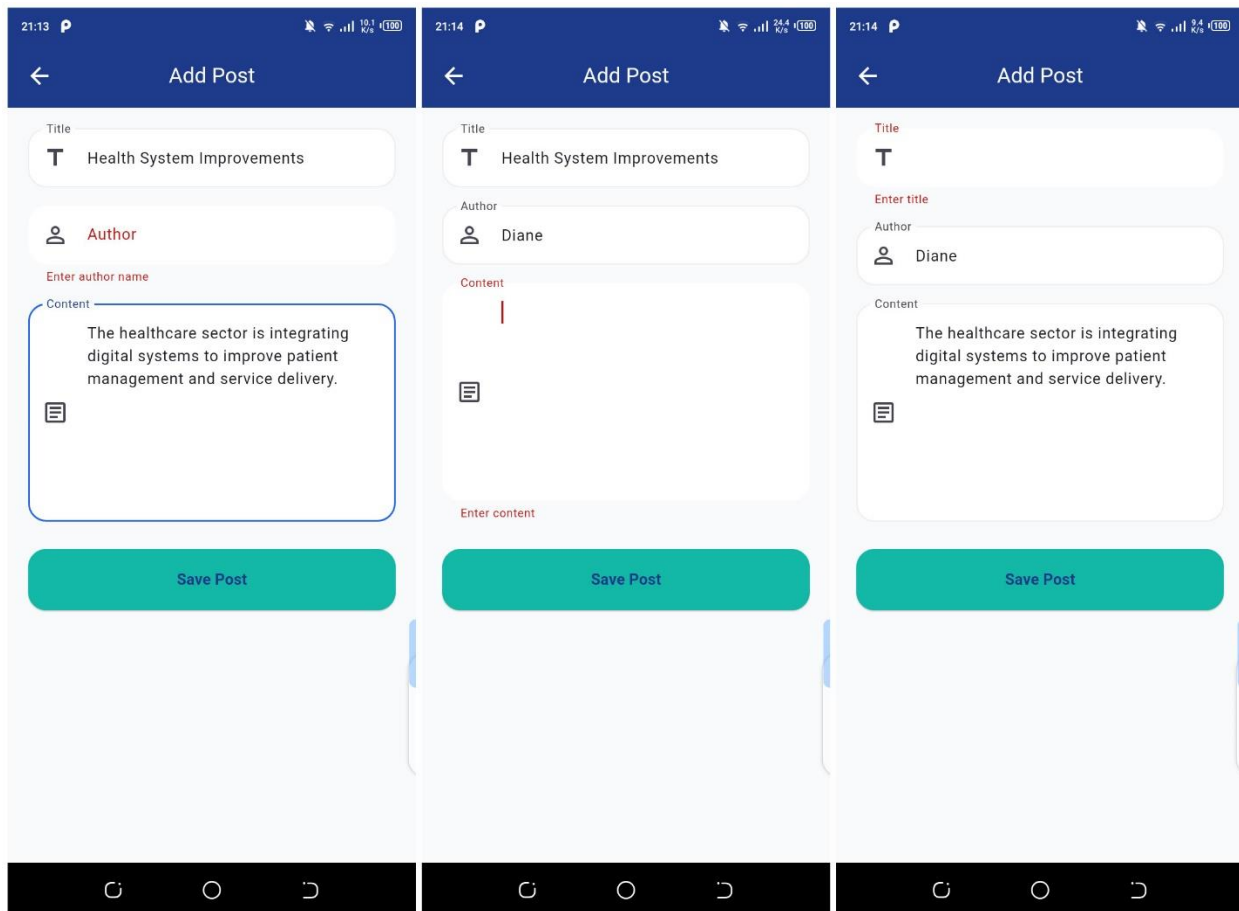
8. Test Scenario 7: Error Handling

Steps:

1. Try saving a post with empty fields

Expected Result:

- Validation errors appear
- Post is not saved



9. Test Scenario 8: Refreshing Data

Steps:

1. Pull down on the home screen (refresh gesture)

Expected Result:

- List refreshes
- No crash occurs

Offline Posts Manager

Education Digitalization Updated

Schools are adopting digital tools and online learning platforms to enhance education accessibil...



Eric Nshimiyimana



Agriculture Innovation

Smart farming techniques using IoT are improving crop yields and helping farmers monitor environme...



Jean Bosco



Rwanda Tech Growth

Rwanda is experiencing rapid growth in the technology sector, with startups and innovation hu...



Aurele



+ New Post

10. Conclusion (For Your Report)

The testing process confirms that the Offline Posts Manager application successfully performs all required CRUD operations using SQLite. Users can add, view, update, and delete posts without internet connectivity. The application handles errors effectively and provides a clean, modern user interface. Screenshots collected during testing demonstrate the functionality of the system.